BET STRUCTURE :
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Etude BA pour l'extension d'une habitation existante en R-1, RDC, Comble
Avenue de dunkerque , 62520 Touquet-Paris-Plage.

Nomenclature des plans structures.

Plan G0 01 : Plan de fondations.

Plan G0 02 : Ph Sous-sol.

Plan G0 03 : Plan rdc.

Plan G0 04 : Ph comble.

Plan G0 05 : Coupe B-B.

Plan G0 06 : Coupe A-A.

Plan G0 07 :Coupe C-C.

Plan G0 08 :Plan de feraillage nappe inf.

Plan G0 09 :Plan de feraillage nappe sup.

Maçonnerie BBM e =20 cm

Batiment existant

Sl 01 : Semelle Isolée de 140 x 140 cm ht
Sl 02 : Semelle Isolée de 160 x 160 cm ht
Pt 01 : Poteau BA 30 x 30 cm ht
Pt 02 : Poteau BA 25 x 25 cm ht
LG 01 : Longrine BA 20x 35 cm ht
LG 01 a :Longrine BA 20x 35 cm ht
LG 02 : Longrine BA 25 x 40 cm ht
LG 03 a : Longrine BA 25 x 40 cm ht
LG 03 b : Longrine BA 25 x 40 cm ht
P 01 a : Poutre BA 25 x 30 cm ht
P 01 b : Poutre BA 25 x 30 cm ht
P 02 a : Poutre BA 25 x 30 cm ht
P 02 b : Poutre BA 20 x 30 cm ht
P 03 : Poutre BA 25 x 30 cm ht
P 04 : Poutre BA 20 x 36 cm ht
P 05 : Poutre BA 20 x 30 cm ht
P 06 a : Poutre BA 30 x 25 cm ht
P 06 b : Poutre BA 30 x 25 cm ht
CH 01: Chainage vertical BA 20 x 20 cm ht
CH 02: Chainage horizontal BA 20 x 20 cm ht

Hypothèses de calculs :

Zone sismique (zone 01)
bâtiment de catégorie I,
la structure n'est pas soumise à une conception parasismique.

Zone de neige : A1

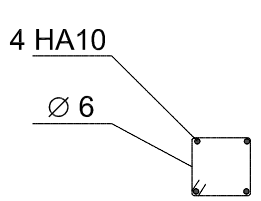
Zone du vent : zone 3

Conformément à l'étude de sols, les fondations sont réalisées par des semelles superficielles.

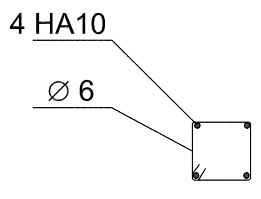
Charges d'exploitation Q = 1.5 Kn / m²

Carellage + chape = 130 kg/m²

Taux de travail du sol est de 1.74 bars



CH 01



CH 02



2 HA10 tout les 25 cm

Principe d'armature béton banché

SI 01

Extension d'une maison existante en R-1; RDC; Comble

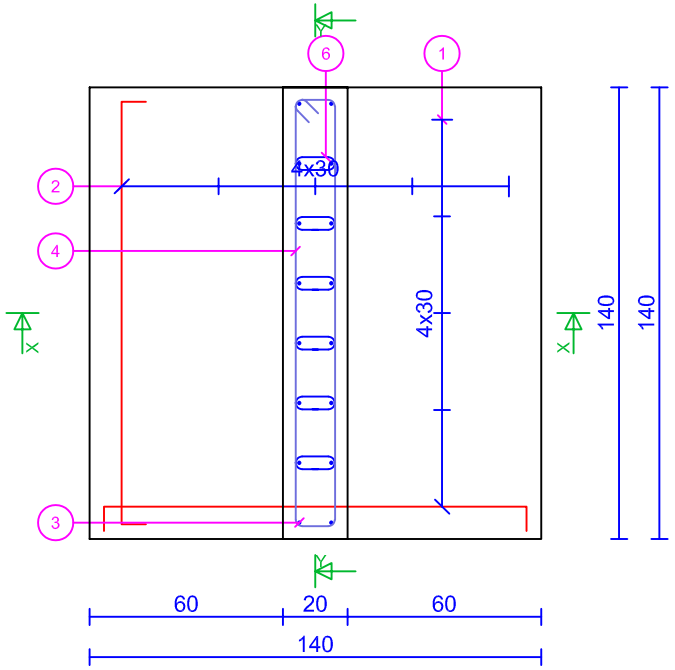
Béton=1.06 m3 Cof=3.8 m²
Acier=19.3 kg d=18.1 kg/m3
Fl=8.8 mm
Densité acier: 7850.00 kg / m3

EB|H=7.5|4.0 cm
EFut=3.5 cm

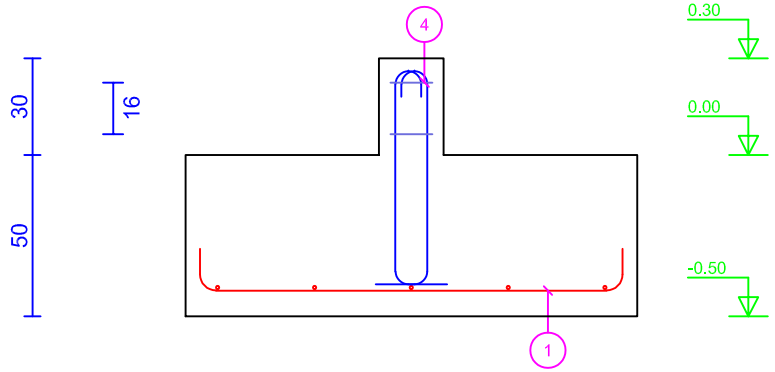
1
1

fck= 25 MPa fyk= 500 MPa | Classe d'exposition: XC2

Elévation
Echelle=1/25



Coupe XX
Echelle=1/25



	Barre	Lg	Forme
1	5HA10	153	13 90° 10 10 13 90° 132
2	5HA10	159	16 90° 10 10 16 90° 132
3	16HA8	93	8 180° 8 8 16 90° 67
4	2HA8	302	13 133
6	12HA8	31	13

Barre	Lg/Poids
HA8	24.6/9.7
HA10	15.6/9.6

SI 02

Extension d'une maison existante en R-1; RDC; Comble

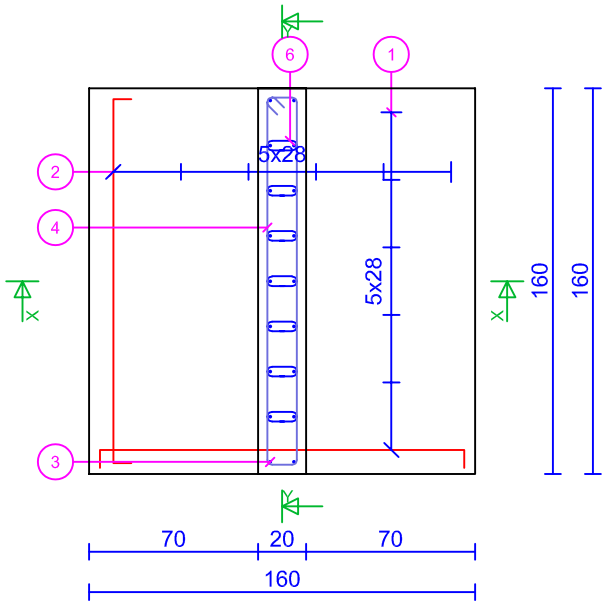
Béton=1.50 m3 Cof=4.6 m²
Acier=24.1 kg d=16.0 kg/m3
Fi=8.9 mm
Densité acier: 7850.00 kg / m3

EB|H=7.5|4.0 cm
EFut=3.5 cm

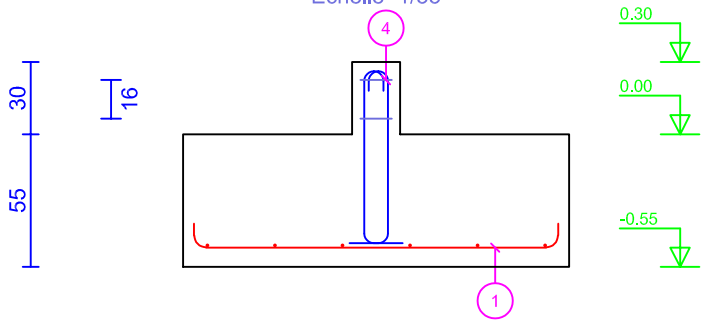
1
1

fck= 25 MPa fyk= 500 MPa | Classe d'exposition: XC2

Elévation
Echelle=1/33



Coupe XX
Echelle=1/33



	Barre	Lg	Forme
1	6HA10	167	11 10 10 11 / 90° 90° / 152
2	6HA10	179	16 10 10 16 / 90° 90° / 152
3	18HA8	98	8 8 8 16 / 180° 90° / 72
4	2HA8	342	13 / 153
6	14HA8	31	13

Barre	Lg/Poids
HA8 HA10	28.8/11.4

Pt 01

Extension d'une maison existante en R-1; RDC; Comble

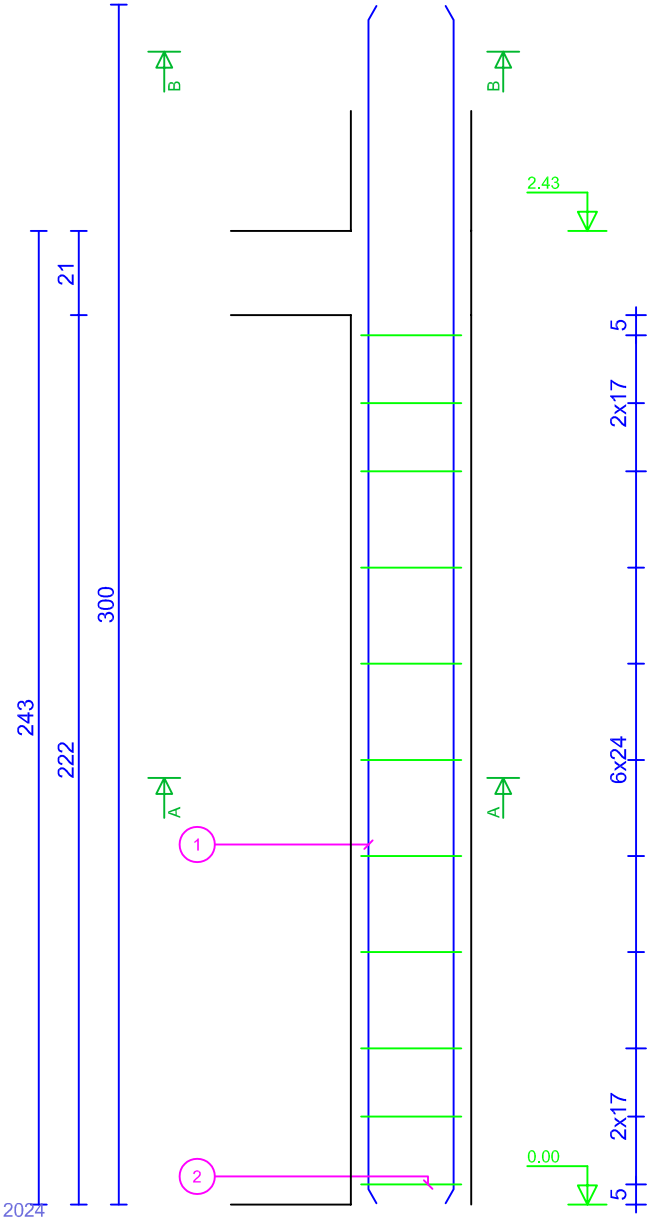
Béton=0.20 m3 Cof=2.7 m²
Acier=17.2 kg d=85.9 kg/m3
Fi=10.8 mm
Densité acier: 7850.00 kg / m3

En=2.5 cm

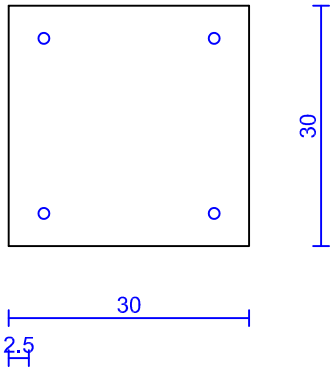
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1

fck= 25 MPa fyk= 500 MPa | Classe d'exposition: XC1

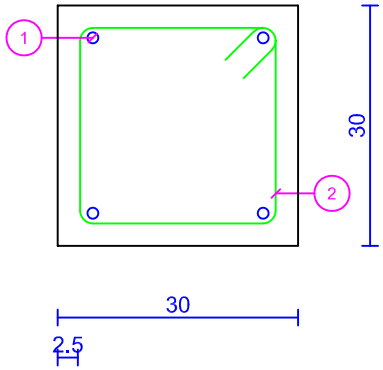
Elévation
Echelle=1/20



Coupe BB Attente sup
Echelle=1/10



Coupe AA Courante
Echelle=1/10



	Barre	Lg	Forme
1	4HA14	300	300
2	11HA6	110	25 / 25

Barre	Lg/Poids
HA6 HA14	12.1/2.7 12.0/14.5

Pt 02

Extension d'une maison existante en R-1; RDC; Comble

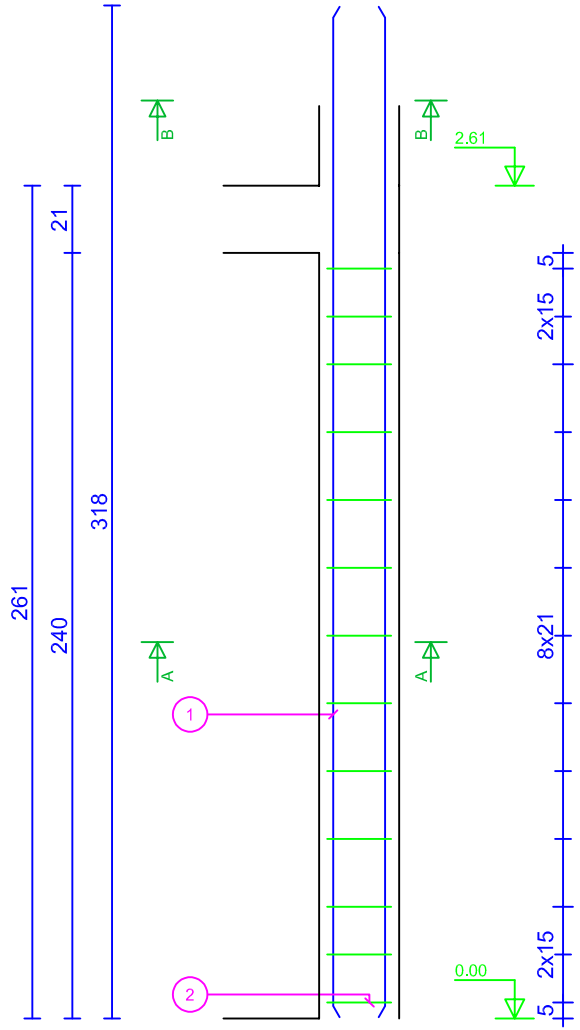
Béton=0.15 m3 Cof=2.4 m²
Acier=17.9 kg d=119.6 kg/m3
Fi=10.9 mm
Densité acier: 7850.00 kg / m3

En=2.5 cm

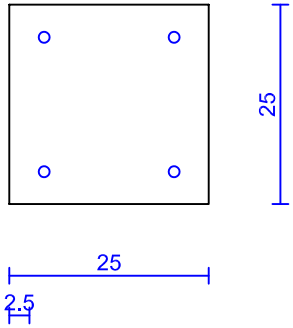
1
1

fck= 25 MPa fyk= 500 MPa | Classe d'exposition: XC1

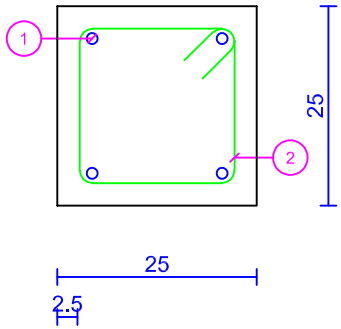
Elévation
Echelle=1/25



Coupe BB Attente sup
Echelle=1/10



Coupe AA Courante
Echelle=1/10



	Barre	Lg	Forme
1	4HA14	318	318
2	13HA6	90	20 / 20

Barre	Lg/Poids
HA6 HA14	11.7/2.6 12.7/15.3

LG 01

Extension d'une maison existante en R-1; RDC; Comble

Béton=0.32 m3
Acier=31.3 kg d=98.0 kg/m3
Fl=8.1 mm Cof=3.7 m²
Densité acier: 7850.00 kg / m3

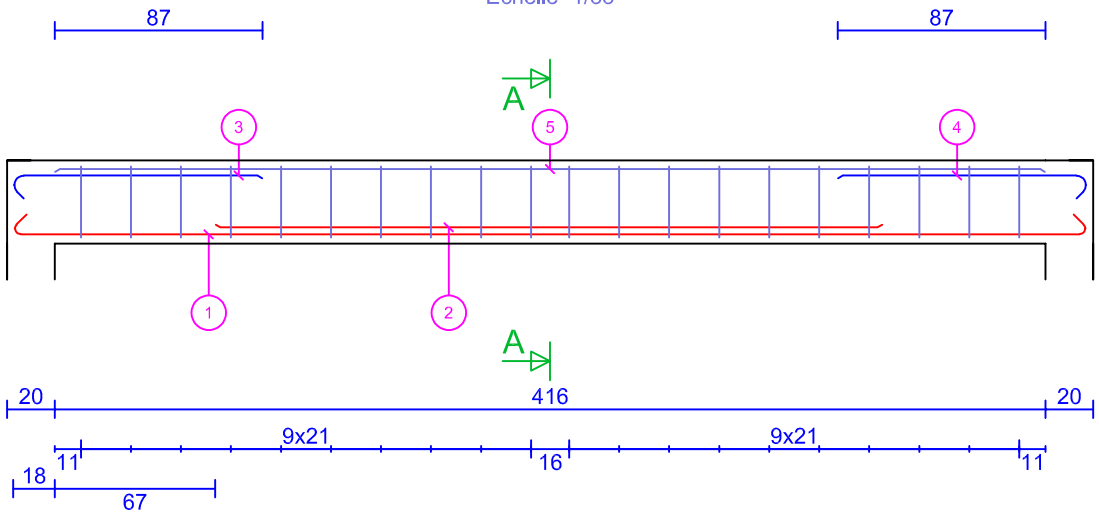
Eb=2.5 cm
Eh=2.5 cm
El=2.5 cm

1
1

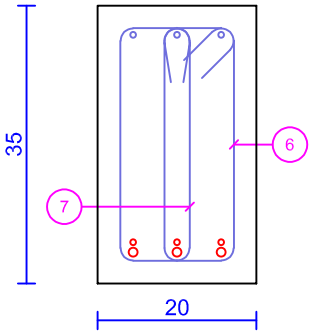
Section : 20 x 35ht

fck= 25 MPa fyk= 500 MPa Classe de ductilité A | Classe d'exposition: XC1

Elévation
Echelle=1/33



Coupe A-A
Echelle=1/10



	Barre	Lg	Forme
1	3HA12	468	9 5 5 9 135° 135° 451
2	3HA8	280	280
3	3HA8	114	8 8 135° 105
4	3HA8	114	8 8 135° 105
5	3HA8	416	416
6	20HA6	99	30 15
7	20HA6	78	30

Barre	Lg/Poids
HA6	35.5/7.9
HA8	27.7/10.9
HA12	14.0/12.5

LG 02

Extension d'une maison existante en R-1; RDC; Comble

Béton=0.38 m3
Acier=19.4 kg d=51.1 kg/m3
Fl=7.1 mm Cof=3.5 m²
Densité acier: 7850.00 kg / m3

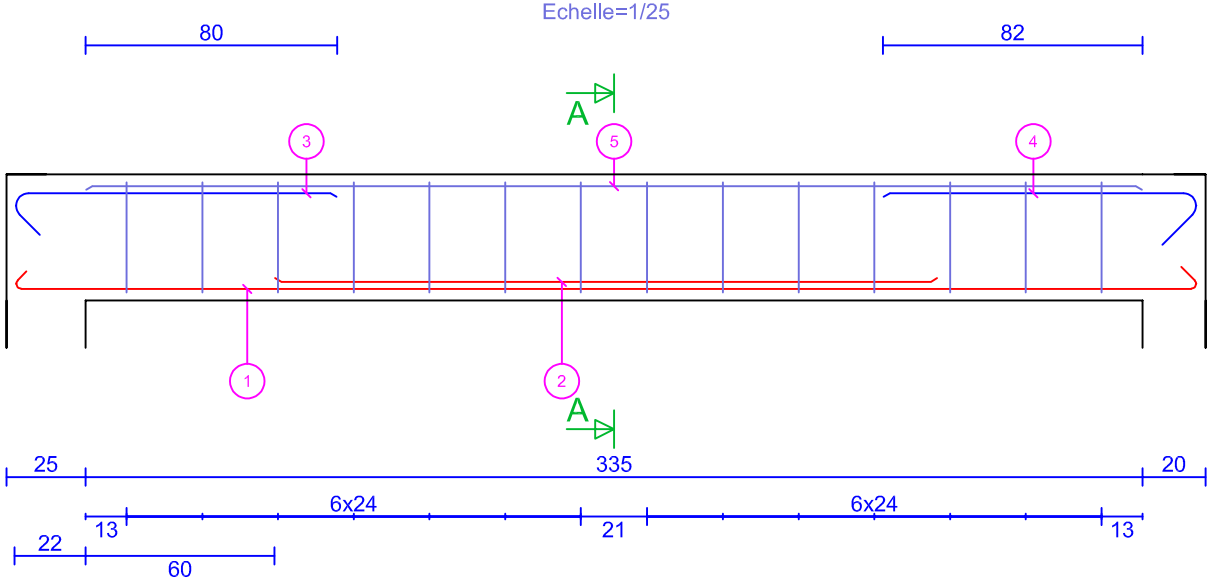
Eb=2.5 cm
Eh=2.5 cm
El=2.5 cm

1
1

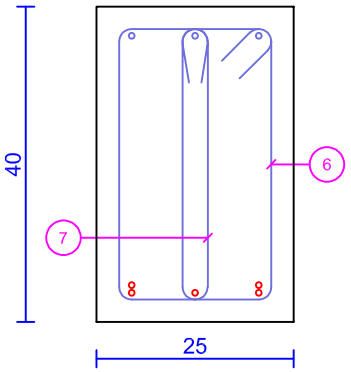
Section : 25 x 40ht

fck= 25 MPa fyk= 500 MPa Classe de ductilité A | Classe d'exposition: XC1

Elévation
Echelle=1/25



Coupe A-A
Echelle=1/10



	Barre	Lg	Forme
1	3HA8	388	6 3 8 3 135° 135° 375
2	2HA8	210	210
3	3HA8	116	13 8 135° 102
4	3HA8	118	18 135° 8 100
5	3HA8	335	335
6	14HA6	119	35 20
7	14HA6	88	35

Barre	Lg/Poids
HA6	29.0/6.4
HA8	32.9/13.0

LG 03 a

Extension d'une maison existante en R-1; RDC; Comble

Béton=0.42 m3
Acier=28.4 kg d=67.7 kg/m3
Fi=12.1 mm Cof=2.9 m²
Densité acier: 7850.00 kg / m3

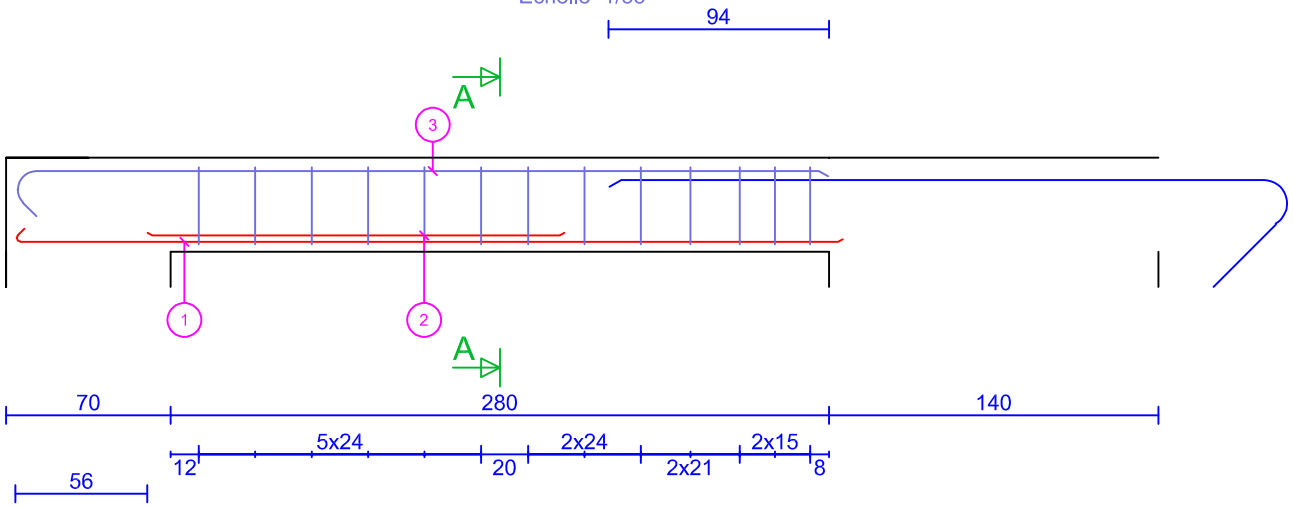
Eb=3.0 cm
Eh=4.0 cm
El=4.0 cm

1
2

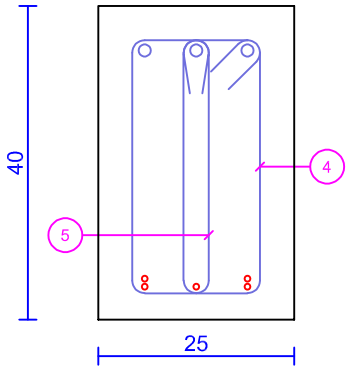
Section : 25 x 40ht

fck= 25 MPa fyk= 500 MPa Classe de ductilité A | Classe d'exposition: XC1

Elévation
Echelle=1/33



Coupe A-A
Echelle=1/10



	Barre	Lg	Forme
1	3HA8	358	6 / 3 / 135° / 352
2	2HA8	178	178
3	3HA16	364	16 / 17 / 135° / 346
4	13HA6	109	33 / 17
5	13HA6	84	33

Barre	Lg/Poids
HA6	25.1/5.6
HA8	14.3/5.6
HA16	10.9/17.2

LG 03 b

Extension d'une maison existante en R-1: RDC: Comble

Béton=0.13 m3
Acier=31.8 kg d=244.6 kg/m3
Fl=13.7 mm Cof=0.6 m²
Densité acier: 7850.00 kg / m3

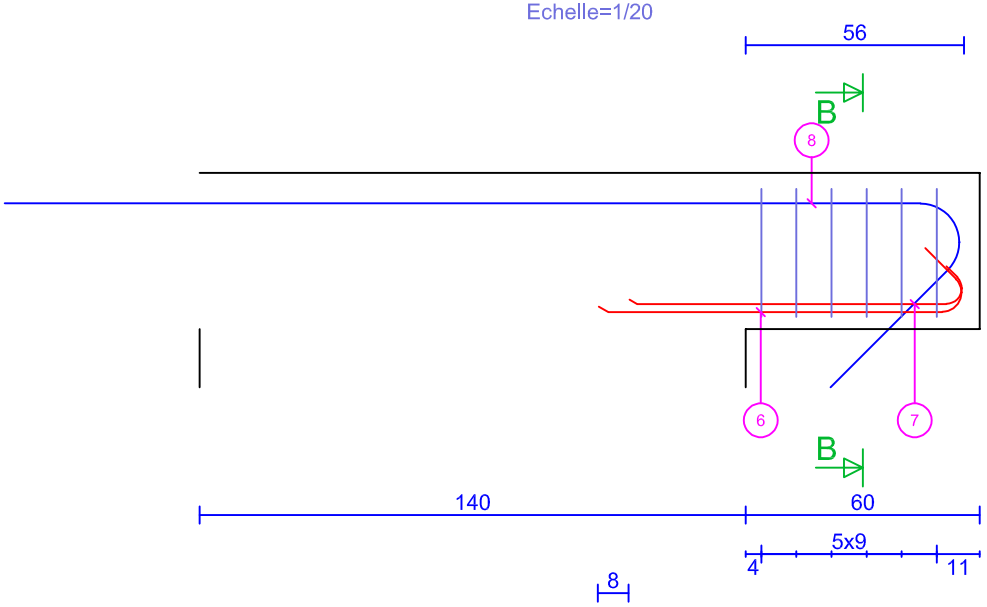
Eb=3.0 cm
Eh=4.0 cm
El=4.0 cm

2
2

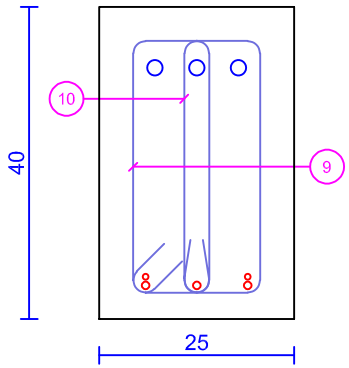
Section : 25 x 40ht

fck= 25 MPa fyk= 500 MPa Classe de ductilité A | Classe d'exposition: XC1

Elévation
Echelle=1/20



Coupe B-B
Echelle=1/10



	Barre	Lg	Forme
6	3HA10	111	10 / 17 / 94 / 135°
7	2HA8	95	8 / 8 / 86 / 135°
8	3HA20	357	66 / 20 / 135° / 290
9	6HA6	109	33 / 17
10	6HA6	84	33

Barre	Lg/Poids
HA6	11.6/2.6
HA8	1.9/0.7
HA10	3.3/2.1
HA20	10.7/26.4

P01 a

Extension d'une maison existante en R-1; RDC; Comble

Béton=0.43 m3
Acier=34.8 kg d=80.9 kg/m3
Fi=7.5 mm Cof=4.5 m²
Densité acier: 7850.00 kg / m3

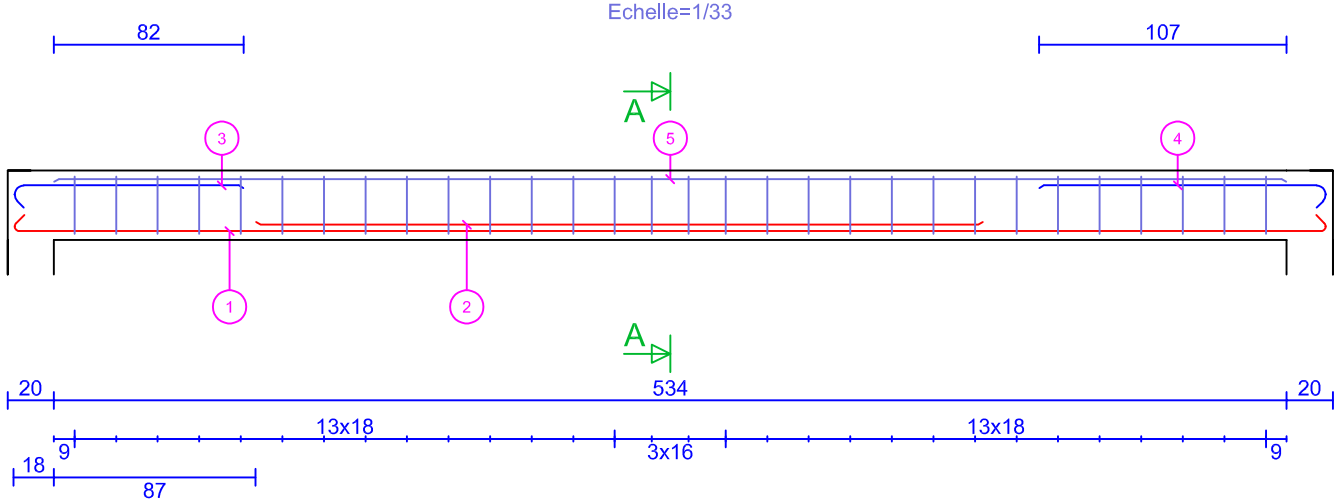
Eb=2.5 cm
Eh=2.5 cm
El=2.5 cm

1
1

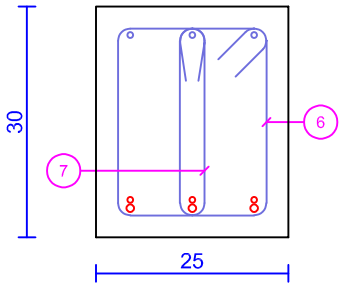
Section : 25 x 30ht

fck= 25 MPa fyk= 500 MPa Classe de ductilité A | Classe d'exposition: XC1

Elévation
Echelle=1/33



Coupe A-A
Echelle=1/10



	Barre	Lg	Forme
1	3HA10	583	135° 4 4 135° 569
2	3HA8	315	315
3	3HA8	109	8 8 8 135° 100
4	3HA8	134	8 8 135° 8 125
5	3HA8	534	534
6	30HA6	99	25 20
7	30HA6	68	25

Barre	Lg/Poids
HA6	50.2/11.1
HA8	32.7/12.9
HA10	17.5/10.8

P01 b

Extension d'une maison existante en R-1; RDC; Comble

Béton=0.17 m3
Acier=12.9 kg d=76.0 kg/m3
Fl=7.2 mm Cof=1.6 m²
Densité acier: 7850.00 kg / m3

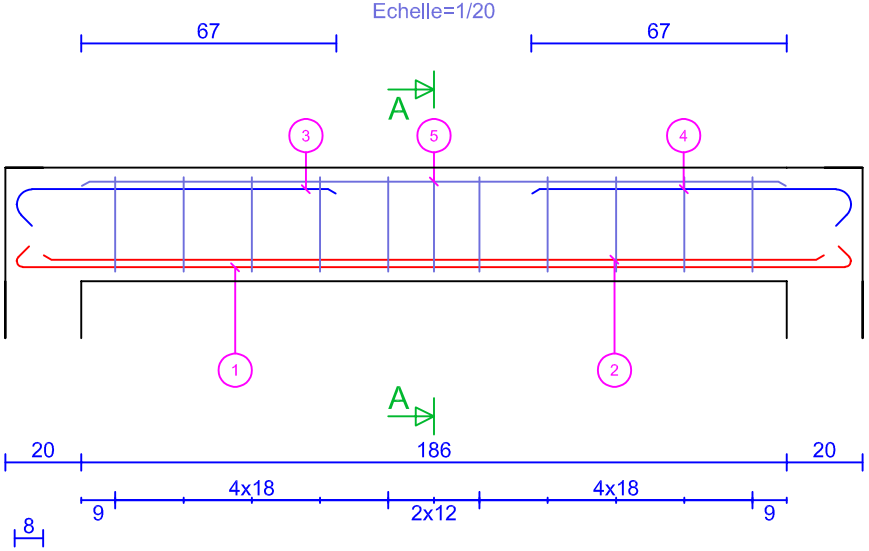
Eb=2.5 cm
Eh=2.5 cm
El=2.5 cm

1
1

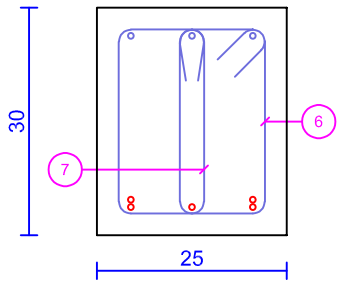
Section : 25 x 30ht

fck= 25 MPa fyk= 500 MPa Classe de ductilité A | Classe d'exposition: XC1

Elévation
Echelle=1/20



Coupe A-A
Echelle=1/10



	Barre	Lg	Forme
1	3HA8	232	6 6 3 3 6 135° 135° 221
2	2HA8	206	206
3	3HA8	94	8 8 8 135° 85
4	3HA8	94	8 8 8 135° 85
5	3HA8	186	186
6	11HA6	99	25 20
7	11HA6	68	25

Barre	Lg/Poids
HA6	18.4/4.1
HA8	22.3/8.8

P02 a

Extension d'une maison existante en R-1; RDC; Comble

Béton=0.43 m3
Acier=42.1 kg d=97.7 kg/m3
Fi=8.2 mm Cof=4.5 m²
Densité acier: 7850.00 kg / m3

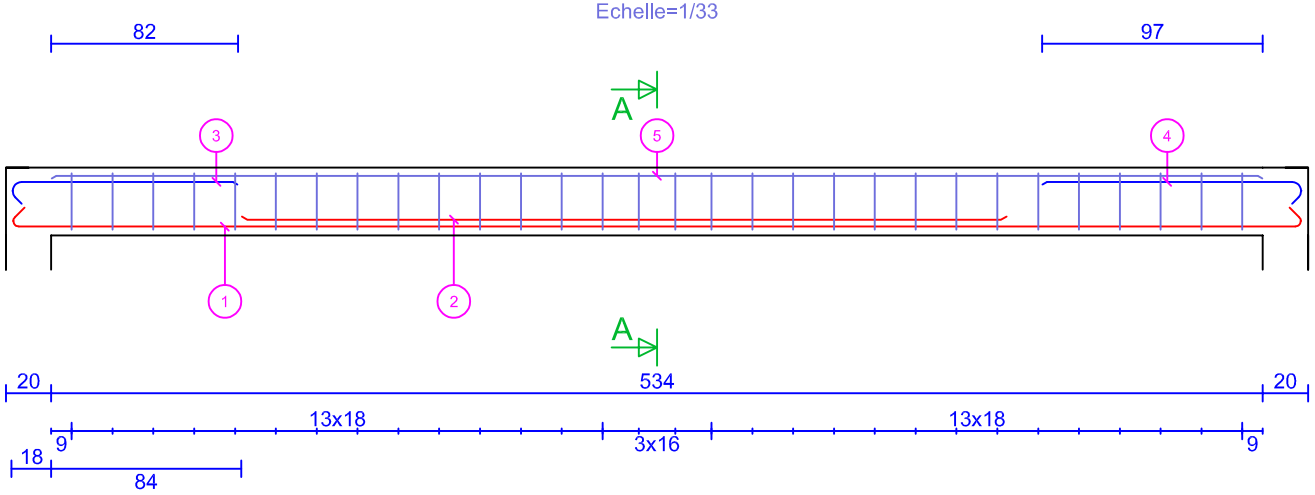
Eb=2.5 cm
Eh=2.5 cm
El=2.5 cm

1
1

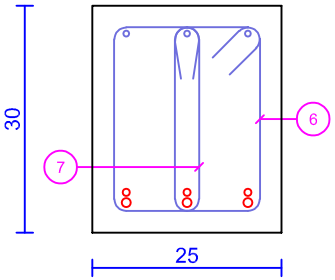
Section : 25 x 30ht

fck= 25 MPa fyk= 500 MPa Classe de ductilité A | Classe d'exposition: XC1

Elévation
Echelle=1/33



Coupe A-A
Echelle=1/10



	Barre	Lg	Forme
1	3HA12	586	135° 5 5 135° 569
2	3HA10	338	338
3	3HA8	109	8 8 135° 100
4	3HA8	124	135° 8 8 115
5	3HA8	534	534
6	30HA6	99	25 20
7	30HA6	68	25

Barre	Lg/Poids
HA6	50.2/11.1
HA8	23.0/9.1
HA10	10.1/6.2
HA12	17.6/15.6

P02 b

Extension d'une maison existante en R-1; RDC; Comble

Béton=0.15 m3
Acier=12.7 kg d=82.7 kg/m3
Fi=7.1 mm Cof=1.7 m²
Densité acier: 7850.00 kg / m3

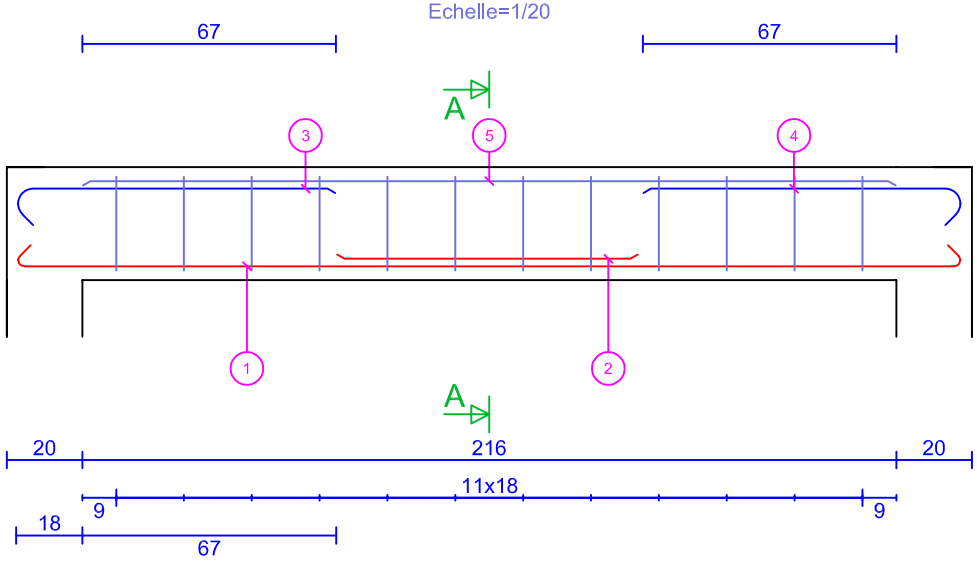
Eb=2.5 cm
Eh=2.5 cm
El=2.5 cm

1
1

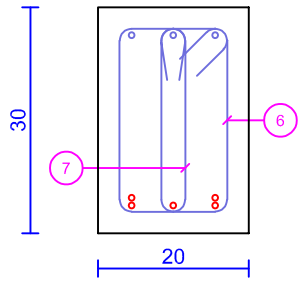
Section : 20 x 30ht

fck= 25 MPa fyk= 500 MPa Classe de ductilité A | Classe d'exposition: XC1

Elévation
Echelle=1/20



Coupe A-A
Echelle=1/10



	Barre	Lg	Forme
1	3HA8	262	6 6 3 3 6 6 135° 135° 251
2	2HA8	80	80
3	3HA8	94	8 8 135° 85
4	3HA8	94	8 8 135° 85
5	3HA8	216	216
6	12HA6	89	25 15
7	12HA6	68	25

Barre	Lg/Poids
HA6 HA8	18.9/4.2 21.6/8.5

P03

Extension d'une maison existante en R-1; RDC; Comble

Béton=0.26 m3
Acier=32.3 kg d=122.5 kg/m3
Fl=8.3 mm Cof=3.2 m²
Densité acier: 7850.00 kg / m3

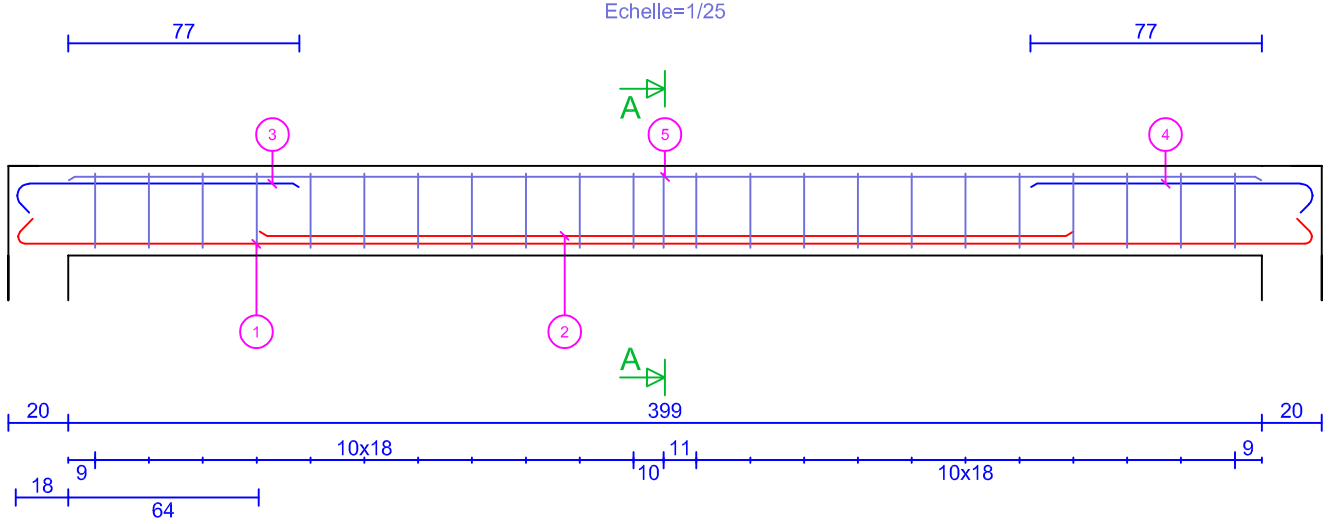
Eb=2.5 cm
Eh=2.5 cm
El=2.5 cm

1
1

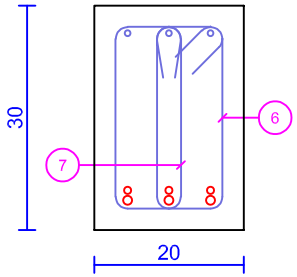
Section : 20 x 30ht

fck= 25 MPa fyk= 500 MPa Classe de ductilité A | Classe d'exposition: XC1

Elévation
Echelle=1/25



Coupe A-A
Echelle=1/10



	Barre	Lg	Forme
1	3HA12	451	9 5 5 9 135° 135° 434
2	3HA10	273	273
3	3HA8	104	8 8 95 135°
4	3HA8	104	8 8 135° 95
5	3HA8	399	399
6	23HA6	89	25 15
7	23HA6	68	25

Barre	Lg/Poids
HA6	36.2/8.0
HA8	18.2/7.2
HA10	8.2/5.0
HA12	13.5/12.0

P04

Extension d'une maison existante en R-1; RDC; Comble

Béton=0.41 m3
Acier=45.8 kg d=110.7 kg/m3
Fi=8.6 mm Cof=4.9 m²
Densité acier: 7850.00 kg / m3

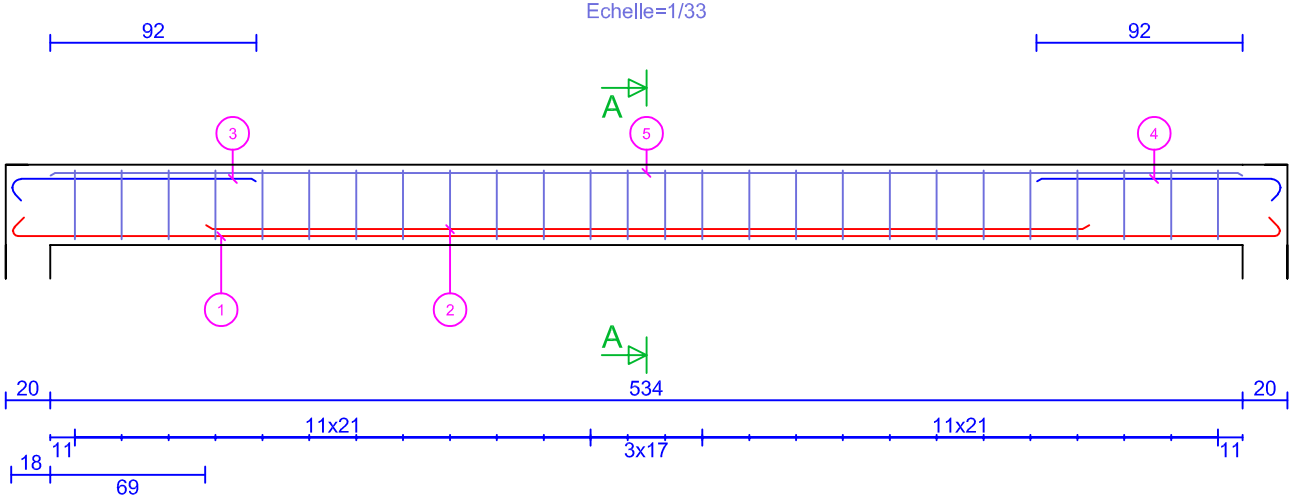
Eb=2.5 cm
Eh=2.5 cm
El=2.5 cm

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1

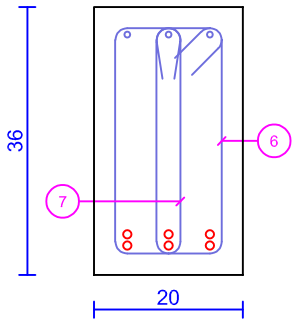
Section : 20 x 36ht

fck= 25 MPa fyk= 500 MPa Classe de ductilité A | Classe d'exposition: XC1

Elévation
Echelle=1/33



Coupe A-A
Echelle=1/10



	Barre	Lg	Forme
1	3HA12	586	9 ⌀5 5 ⌀6 135° 569 135°
2	3HA12	396	396
3	3HA8	119	8 ⌀8 110 135°
4	3HA8	119	8 135° ⌀8 110
5	3HA8	534	534
6	26HA6	101	31 15
7	26HA6	80	31

Barre	Lg/Poids
HA6 HA8 HA12	47.1/10.5 23.1/9.1

P05

Extension d'une maison existante en R-1; RDC; Comble

Béton=0.23 m3
Acier=38,5 kg d=169,0 kg/m3
Fi=9.7 mm Cof=2.7 m²
Densité acier: 7850.00 kg / m3

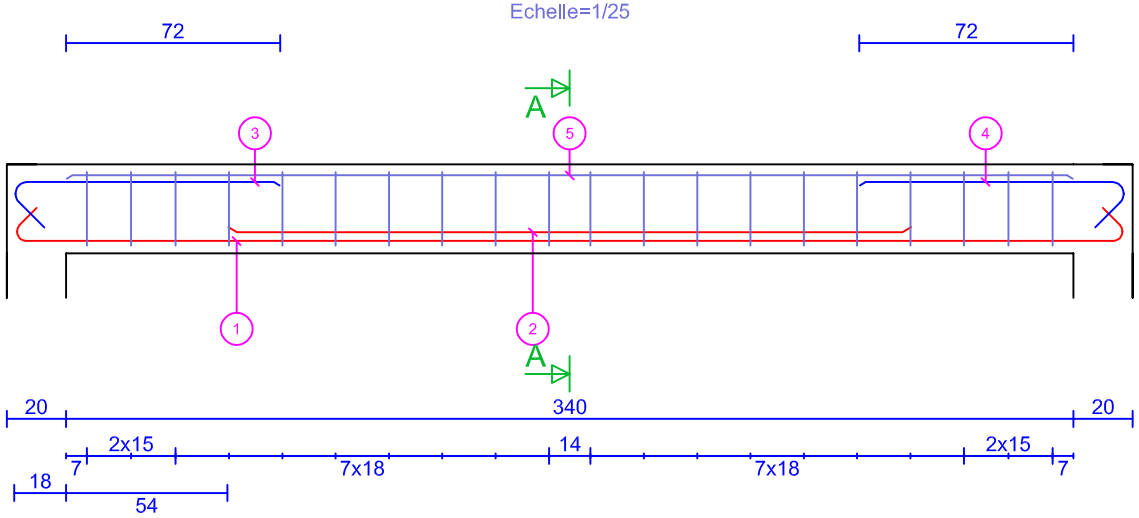
Eb=2.5 cm
Eh=2.5 cm
El=2.5 cm

1
1

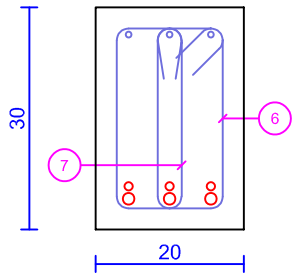
Section : 20 x 30ht

fck= 25 MPa fyk= 500 MPa Classe de ductilité A | Classe d'exposition: XC1

Elévation
Echelle=1/25



Coupe A-A
Echelle=1/10



	Barre	Lg	Forme
1	3HA16	398	12 6 6 12 135° 135° 375
2	3HA12	231	231
3	3HA8	107	16 8 135° 90
4	3HA8	107	16 8 135° 90
5	3HA8	340	340
6	20HA6	89	25 15
7	20HA6	68	25

Barre	Lg/Poids
HA6	31.5/7.0
HA8	16.6/6.6
HA12	6.9/6.2
HA16	11.9/18.8

P06

Extension d'une maison existante en R-1; RDC; Comble

Béton=0.19 m3
Acier=21.6 kg d=112.6 kg/m3
Fl=8.2 mm Cof=1.7 m²
Densité acier: 7850.00 kg / m3

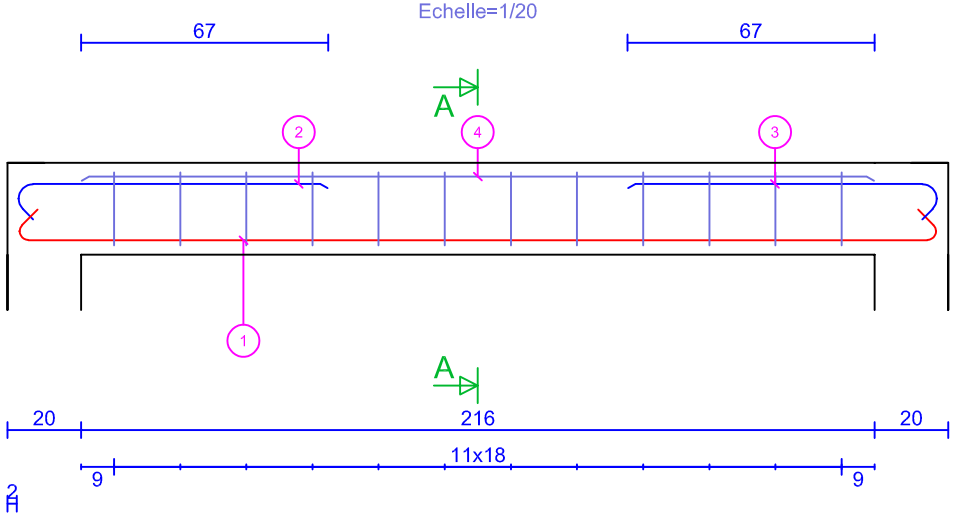
Eb=2.5 cm
Eh=2.5 cm
Et=2.5 cm

1
1

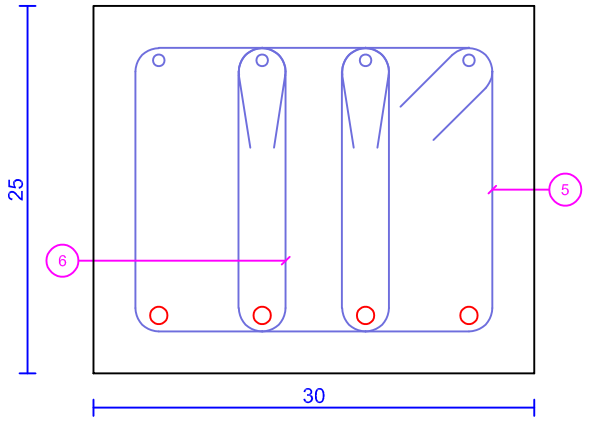
Section : 30 x 25ht

fck= 25 MPa fyk= 500 MPa Classe de ductilité A | Classe d'exposition: XC1

Elévation
Echelle=1/20



Coupe A-A
Echelle=1/5



	Barre	Lg	Forme
1	4HA12	268	9 6 5 5 6 9 135° 135° 251
2	4HA8	94	8 8 135° 85
3	4HA8	94	8 8 135° 85
4	4HA8	216	216
5	12HA6	99	20 25
6	24HA6	58	20

Barre	Lg/Poids
HA6	25.8/5.7
HA8	16.1/6.4
HA12	10.7/9.5